

Jincang (Jin) Li

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PERSONAL SUMMARY

Robotics and AI engineer with 3 years of experience in ROS-based autonomous robot systems, embedded systems, and real-world service robot deployment. Skilled in localization, SLAM, sensor fusion, path planning, obstacle avoidance, computer vision and perception, and multi-sensor integration. Research interests focus on embodied intelligence, including dexterous manipulation, grasp generation, hand-object interaction modelling, reinforcement learning, Vision-Language-Action (VLA) models, and physics-aware robot simulation.

EDUCATION

Western University

London, ON, Canada

Master of Engineering Science (MEngSc) in Electrical and Computer Engineering. GPA: 3.7/4.0 (Grade: A) Sep. 2024 – Sep. 2026

- Key Courses: Robotics Manipulators, Linear Systems and Modern Control, Sensor Networks and Embedded Systems, Topics in Autonomous Robotics, Advanced Digital Control

Dalian Minzu University

Dalian, Liaoning, China

Bachelor of Engineering in Measurement Control Technology and Instruments, GPA: 84.6/100 (Grade: A) Sep. 2018 - Jun. 2022

- Key Courses: Principles of Electric Circuits, Digital Electronic Technology, Analog Electronic Technology, Principles of Automatic Control, The Principle and Interface Technology of Microcomputer

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, Bash

Robotics & ROS: ROS, ROS Navigation Stack, TF/TF2, URDF/Xacro, RViz, ROSBag, Costmap

Localization & Mapping: SLAM, NDT Localization, ICP, AMCL, EKF Sensor Fusion, Wheel Odometry, Occupancy Grid Mapping

Planning & Control: A*, DWA, Global/Local Planning, Obstacle Avoidance, PID Control, Trajectory Tracking, Robot Kinematics

Robot Learning & Manipulation: Dexterous Grasping, 6D Grasp Pose Estimation, MANO Hand Model, Multi-Head Regression, Transformer, Diffusion

Perception & Point Cloud Processing: Point Cloud Processing, PointNet++, Open3D, OpenCV, Surface Sampling, Geometric Constraint Modeling

Deep Learning: PyTorch, Neural Network Training, Multi-Task Loss Design, Data Augmentation, Gradient Clipping

Simulation: Isaac Gym, MuJoCo, Robot Simulation, Physics-Based Grasp Evaluation

Embedded Systems: STM32, Keil, ADC/DAC, Sensor Integration, Bluetooth, Serial Communication

Tools: Linux, Git, CMake, GDB, MQTT, TCP/UDP, Altium Designer, SolidWorks, Multisim

WORK EXPERIENCE

Beijing Zhenji Intelligent Technology Co., Ltd

Seoul, South Korea

ROS Algorithm Engineer

Jun. 2022 - Sep. 2024

- Developed and deployed ROS-based localization, navigation, and autonomous delivery modules for service robots operating in South Korea and multiple cities in China.
- Contributed to a commercial robot delivery system that has been in operation for over 4 years, with more than 800,000 km of cumulative robot travel distance.
- Maintained and optimized the South Korea regional deployment, achieving a robot delivery success rate above 94%.
- Integrated LiDAR, cameras, ultrasonic sensors, wheel odometry, and IMU/attitude sensors for perception, localization, and navigation.
- Implemented localization and navigation algorithms, including odometry fusion, SLAM-based pose estimation, ROS Navigation Stack, global path planning, local obstacle avoidance, and target navigation.
- Developed elevator interaction functions using MQTT/SaaS communication, including elevator scenario handling, button-operation logic, and in-elevator navigation.
- Conducted system testing, ROSBag-based debugging, sensor data analysis, and code optimization to improve navigation robustness and deployment reliability.

Shanghai Yixing Electromechanical Equipment Co., Ltd

Shanghai, China

Intern Embedded Systems Engineer

Feb. 2022 - Jun. 2022

- Developed embedded software modules for the RPD-100L personal radiation dose alarm device, including dose-rate measurement, cumulative dose recording, alarm threshold configuration, data storage, and UI control.
- Implemented radiation data processing, unit auto-switching, filtering, and fast alarm-response logic to improve measurement stability and protection reliability.
- Supported hardware selection, circuit optimization, and product validation for radiation detection, signal conditioning, ADC,

MCU, and low-power management circuits.

Dalian Haoke Technology Co., Ltd

Dalian, Liaoning, China

Intern Software Engineer

Jul. 2019 - Sep. 2019

- Developed embedded software for an electric toothbrush control system, including vibration control, charging control, battery monitoring, and low-battery protection.
- Designed circuit schematics, selected electronic components, created PCB layouts, and implemented driver programs for microcontroller-based hardware control.

PROJECT EXPERIENCE

Dexterous Grasp Generation with Learning-Based 21-Joint Hand Pose Prediction

02.2025-06.2026

- Developed a PointNet++-based dexterous grasp generation framework to predict MANO 21-joint hand poses from object point clouds.
- Designed a multi-head regression network to generate multiple grasp candidates in a single forward pass without iterative sampling.
- Implemented object-level multi-GT supervision with oracle matching and Top-K head selection to model multiple feasible grasps for the same object.
- Introduced geometric constraints, including bone length, bone direction, fingertip diversity, and fingertip-to-object proximity losses, to improve physical plausibility.
- Fitted predicted discrete hand joints to continuous MANO parameters and refined hand-object geometry through surface sampling, contact modeling, and penetration control.

7-Axis Robotic Arm AI-Based Grasp Training Project

01.2024-06.2024

- Built a 7-axis robotic arm grasping platform and integrated hardware/software modules for manipulation experiments.
- Implemented deep learning-based 6D grasp pose estimation for object manipulation in simulated environments.
- Conducted GPU-accelerated grasp training in Isaac Gym to support large-scale parallel simulation and policy evaluation.

ROS-Based Autonomous Driving System

02.2022-06.2022

- Developed a ROS-based autonomous driving system for time-constrained navigation from a start point to a target point in obstacle-filled environments.
- Designed obstacle avoidance and path planning logic to generate collision-free trajectories while minimizing travel time to the target.
- Tuned PID-based motion control for steering and speed tracking, improving vehicle stability during sharp turns, obstacle avoidance, and rapid acceleration/deceleration.
- Recorded and analyzed ROS sensor data and vehicle-state logs to debug localization drift, control delay, obstacle avoidance failures, and path-tracking errors.

Industrial Pipeline Leak Detection System based on IoT

02.2021-02.2022

- Designed an IoT-based pipeline leakage monitoring system for heating pipelines, flanges, and critical pipeline equipment.
- Developed a modular inspection framework combining pipeline-climbing robots, flange leakage detection nodes, ground patrol vehicles, and magnetic pipeline vehicles.
- Implemented sensor data acquisition, wireless transmission, backend data processing, and alarm logic for leakage detection.
- Applied image/sensor-based analysis and data-driven processing methods to identify abnormal leakage patterns and support maintenance decisions.

PATENTS & PUBLICATIONS

Co-inventor of 10 patent applications related to pipeline inspection robots, IoT-based leakage detection, and intelligent monitoring systems.

Selected patents(2 out of 10):

- Patent (CN202010364709.5): "Internet of Things Pipeline Inspection Robot and Intelligent Leakage Detection System," Applied Aug. 10, 2020.
- Patent (CN202010364710.8): "AI-Based Pipeline Inspection System for Curve-Passable Leakage Detection, Positioning, and Monitoring," Applied Aug. 13, 2020.

HONORS & AWARDS

- Recipient, Fred Groch Graduate Bursary in Engineering, Western University, (\$5,000). 2025–2026
- Awarded five undergraduate scholarships for academic excellence, totalling RMB 28,000, Dalian Minzu University. 2018-2022

Received 25 awards in total, including 9 national-level awards, 15 provincial-level awards, and 1 municipal-level award.

Selected national-level awards(3 out of 25):

- Excellent Award, The 15th National College Student Transportation Technology Competition Nov. 2020
- Second Prize, National College Student Electronic Design Competition Oct. 2020
- Third Prize, China Engineering Robotics Competition Dec. 2020